

Weighting Scheme: Inverse Volatility (Naive Risk Parity)

MetaFVG Mean-Reversion Portfolio – 14 Instruments

August 1, 2026

1 Scheme Description

The portfolio blends 14 fundamentally diversified instruments (7 FX majors/crosses, 4 equity indices, and 3 additional FX pairs spanning different central banks) trading a Spearman-gated MetaFVG mean-reversion strategy. The current production blend uses equal notional weights (1/14 each). This report tests whether an **inverse-volatility** (naive risk-parity) weighting scheme improves realized diversification relative to equal weighting.

Each instrument’s own standalone daily-return volatility σ_i is computed over its own non-NaN trading days (days with no closed trade for that instrument are excluded from that instrument’s own vol estimate). Weights are set inversely proportional to σ_i , so instruments with lower standalone volatility receive higher weight – equalizing each instrument’s naive volatility contribution while completely ignoring the correlation structure between instruments:

$$w_i = \frac{1/\sigma_i}{\sum_{j=1}^{14} 1/\sigma_j}, \quad \sigma_i = \text{std}(\{r_{i,t} : r_{i,t} \neq \text{NaN}\}) \quad (1)$$

On any given day, only instruments with a non-NaN return that day contribute; the scheme’s weights are renormalized to sum to 1 over the subset of instruments trading that day before the weighted sum is taken. Days on which *all* 14 instruments are NaN (no trade closed anywhere) are dropped from the blended series.

2 Weights

Low-volatility FX pairs (USDUSD, USDCAD, EURUSD, GBPUSD) receive up to 1.9x the equal-weight allocation, while the higher-volatility equity indices (JP225, HK50) and equity/EM instruments are cut to roughly 0.4-0.6x equal weight.

3 Correlation Impact

The diversification ratio is defined as $DR = \frac{\sum_i w_i \sigma_i}{\sigma_{\text{portfolio}}}$, where $\sigma_{\text{portfolio}}$ is the standard deviation of the final blended daily-return series. Higher DR indicates more realized diversification benefit from combining the instruments.

Table 1: Inverse-volatility weights, sorted descending. Ratio column is $w_i/(1/14)$, i.e. the multiple of the equal-weight allocation.

Symbol	Weight	Ratio to 1/14	σ_i (daily)
USDSGD	0.1351	1.8917	0.000 020
USDCAD	0.1120	1.5679	0.000 024
EURUSD	0.1050	1.4705	0.000 026
GBPUSD	0.0890	1.2467	0.000 030
EURNOK	0.0839	1.1750	0.000 032
USDCHF	0.0834	1.1670	0.000 032
USDJPY	0.0709	0.9930	0.000 038
AUDUSD	0.0654	0.9157	0.000 041
NZDUSD	0.0571	0.7996	0.000 047
USDZAR	0.0495	0.6937	0.000 054
GER40	0.0450	0.6304	0.000 060
US500	0.0431	0.6037	0.000 063
HK50	0.0311	0.4357	0.000 087
JP225	0.0293	0.4095	0.000 092

Table 2: Weighted average pairwise correlation and diversification ratio, equal-weight baseline vs. inverse-volatility scheme.

Metric	Equal Weight	Inverse Vol
Weighted avg. pairwise correlation	0.015 615	0.015 947
Diversification ratio (DR)	2.080 922	1.808 789

4 Results

5 Takeaway

The inverse-volatility scheme did not improve realized diversification: the weighted average pairwise correlation was essentially unchanged (0.01595 vs. 0.01561, marginally *worse*), and the diversification ratio actually fell from 2.08 to 1.81. This makes sense in hindsight – inverse-vol weighting ignores the correlation matrix entirely and simply concentrates capital in the lowest-volatility instruments (USDSGD, USDCAD, EURUSD), which does nothing to reduce co-movement and instead reduces the effective breadth of the book. Sharpe dropped substantially, from 0.518 to 0.321 (-38%), along with Sortino, Calmar, and recovery factor all deteriorating in tandem. Equal weighting remains the better scheme of the two tested here; naive risk parity is not a good fit for a portfolio that was already assembled for fundamental (not volatility-based) diversification.

Table 3: Quantstats risk/return metrics, equal-weight baseline vs. inverse-volatility scheme.

Metric	Equal Weight	Inverse Vol
Sharpe	0.5179	0.3214
Sortino	0.8170	0.5037
Calmar	0.3430	0.2001
CAGR	0.0002	0.0001
Max drawdown (%)	−5.3252	−5.3252
Volatility (% , ann.)	3.5276	3.3172
Tail ratio	1.2387	1.1027
Ulcer index	0.0002	0.0002
Kelly criterion	0.0668	0.0471
VaR (%)	−0.0036	−0.0034
CVaR (%)	−0.0072	−0.0076
Skew	1.5448	1.9403
Recovery factor	2.2601	1.3186
Win rate	0.3089	0.3016
Best day	0.000 291	0.000 291
Worst day	−0.000 239	−0.000 239
Total return	0.001 204	0.000 702